

## BIOTECHNOLOGY AND ITS APPLICATION

1. A transgenic food crop, which may help in solving the problem of night blindness in developing countries is :
  - (1) Bt soyabean
  - (2) Golden rice
  - (3) Flavr savr tomatoes
  - (4) Starlink maize
2. Bacterium *Bacillus thuringiensis* is widely used in contemporary biology as
  - (1) Source of industrial enzyme
  - (2) Indicator of water pollution
  - (3) Insecticide
  - (4) Agent for production of dairy products.
3. Bt toxin is produced by *Bacillus thuringiensis* which is :
  - (1) Bacterium
  - (2) Protozoa
  - (3) Fungus
  - (4) Virus
4. GEAC stands for :
  - (1) Gene evaluation approval committee
  - (2) Genetic engineering approval committee
  - (3) Genetic engineering applied committee
  - (4) Gene enhancement approval committee
5. Conventional methods to diagnose a disease are :
  - (1) Serum and urine analysis
  - (2) PCR
  - (3) ELISA
  - (4) All of the above
6. The first transgenic cow, which produced human protein enriched milk was named :
  - (1) Andy
  - (2) Dolly
  - (3) Rosie
  - (4) Dumpy
7. Milk of transgenic cow 'Rosie' contains a substance that was nutritionally more balanced product for human babies is :
  - (1)  $\alpha$ -lactalbumin
  - (2)  $\beta$ -lactalbumin
  - (3)  $\gamma$ -lactalbumin
  - (4)  $\delta$ -lactalbumin
8. Which peptide is not present in the mature insulin and is removed during maturation into insulin?
  - (1) A-peptide
  - (2) B-peptide
  - (3) C-peptide
  - (4) Both (1) and (2)
9. Two polypeptide chains of insulin are linked together by :
  - (1) disulphide bonds
  - (2) hydrogen bonds
  - (3) Phosphodiester bonds
  - (4) Glycosidic bonds
10. 'Flavr Savr' is a transgenic variety of :
  - (1) Potato
  - (2) Tomato
  - (3) Soyabean
  - (4) Rice
11. Golden rice is enriched in :
  - (1) Vitamin C
  - (2) Vitamin D
  - (3) Vitamin A
  - (4) Vitamin E
12. For the control of the cotton bollworms, which one of the genes is useful ?
  - (1) Cry 1 Ac
  - (2) Cry 1 Ab
  - (3) Cry 1 Ad
  - (4) All of these
13. Process involving silencing of a specific mRNA due to a complementary dsRNA molecule is called :
  - (1) Transcription
  - (2) RNA interference
  - (3) DNA interference
  - (4) None of these
14. *Meloidogyne incognita* which infects the roots of tobacco plants causing a great reduction in yield is a :
  - (1) Nematode
  - (2) Bacterium
  - (3) Virus
  - (4) Alga
15. Toxin present in *Bacillus thuringiensis* does not kill the bacterium because it is inactive form what makes it active inside the insect ?
  - (1) the alkaline pH of the gut, which solubilises the crystals
  - (2) the acid pH of the gut
  - (3) the neutral pH of the gut
  - (4) All of the above

- 16.** Crystals of Bt toxin produced by some bacteria do not kill the bacteria themselves because :
- (1) bacteria are resistant to the toxin
  - (2) toxin is immature
  - (3) toxin is inactive
  - (4) bacteria enclose toxin in a special sac
- 17.** Which one of the following statements about genetically engineered insulin is incorrect.
- (1) *E.coli* is used for producing humulin
  - (2) Chains A, B were produced separately
  - (3) Eli lily company prepared it for first time
  - (4) Genetically engineered insulin has C-peptide
- 18.** Infection by pathogen can be detected by the presence of antigens or by detecting the antibodies synthesised against the pathogen, on this principle a test is based which is ?
- (1) PCR
  - (2) ELISA
  - (3) Both (1) and (2)
  - (4) None of the above
- 19.** Indian parliament recently cleared, which amendment of the Indian patents bill,
- (1) First amendment      (2) Second amendment
  - (3) Third amendment      (4) Fourth amendment
- 20.** How many documented varieties of basmati rice distinct for its unique aroma and flavour are grown in India?
- (1) 27 varieties      (2) 25 varieties
  - (3) 28 varieties      (4) 26 varieties
- 21.** Over 95 percent of all existing transgenic animals are :
- (1) Pigs      (2) Cows
  - (3) Fish      (4) Mice
- 22.** The organisation set up for making decisions regarding the validity of GM research and the safety of introducing GM organism for public services is :
- (1) Genetic engineering approval committee
  - (2) Genetic engineering advanced company
  - (3) Genetic engineering applied committee
  - (4) None of these
- 23.** Use of bio-resources by multinational companies and other organisations without proper authorisation from the countries and people concerned without compensatory payment is called :
- (1) Biotheft
  - (2) Biopatent
  - (3) Biopiracy
  - (4) None of the above
- 24.** Which of the following plants is genetically modified for improved nutritional value of food?
- (1) Potato      (2) Wheat
  - (3) Rice      (4) Maize
- 25.** Which animal is being used to test the safety of polio vaccine?
- (1) Transgenic mice
  - (2) Transgenic pig
  - (3) Transgenic cow
  - (4) Transgenic cat
- 26.** Cry1 Ab gene produces proteins which control?
- (1) Bollworms
  - (2) Corn borer
  - (3) Both (1) and (2)
  - (4) None of the above
- 27.** Nematode-specific genes were introduced into the host plant (tobacco plant) by using which vector?
- (1) Plasmid vector
  - (2) Cosmid vector
  - (3) Bacteriophage vector
  - (4) BAC
- 28.** Which one of the following statements are true regarding genetic modifications ?
- (1) Genetic modifications reduced reliance on chemical pesticides
  - (2) Genetic modifications has enhanced nutritional value of food.
  - (3) Genetic modifications made crops more tolerant to abiotic stresses.
  - (4) All are correct

- 29.** Critical research areas of biotechnology are :
- (1) providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.
  - (2) Creating optimal conditions through engineering for a catalyst to act.
  - (3) Down stream processing technologies to purify the protein/organic compound.
  - (4) All the above
- 30.** Bacterium genetically engineered for cleaning oil spills is :
- (1) *Escherichia coli*
  - (2) *Pseudomonas putida*
  - (3) *Salmonella typhimurium*
  - (4) *Agrobacterium tumefaciens*
- 31.** Bacterium which is known as 'Super bug' is :
- (1) *Pseudomonas putida*
  - (2) *Salmonella*
  - (3) *Escherichia*
  - (4) *Agrobacterium*
- 32.** Animals those have had their DNA manipulated to possess and express an extra (foreign) gene are known as :
- (1) Transgenic animals
  - (2) Genetically modified animals
  - (3) Both (1) and (2)
  - (4) None of the above
- 33.** When cut by the restriction enzyme, the DNA fragments can be joined together using :
- (1) DNA polymerase      (2) DNA ligase
  - (3) Alkaline phosphatase      (4) DNA gyrase
- 34.** Genetically engineered human insulin is made in
- (1) Fungus      (2) Protista
  - (3) Plants      (4) Bacterium
- 35.** Genetically engineered bacteria are being used in commercial production of :
- (1) melatonin      (2) testosterone
  - (3) thyroxine      (4) human insulin
- 36.** Insulin consists of two short polypeptide chains, which are linked together by?
- (1) Sulphide bridges      (2) Peptide bridges
  - (3) Chloride bridges      (4) Disulphide bridges
- 37.** *Meloidogyne incognitia* infects the root of which plant ?
- (1) Potato      (2) Soyabean
  - (3) Tobacco      (4) Tomato
- 38.** Genetics modified crops (GMC) are useful in agriculture because :
- (1) They are more tolerant to abiotic stresses
  - (2) They increase reliance on chemical pesticide
  - (3) They have reduced nutritional value
  - (4) All the above
- 39.** The protein encoded by the gene cryI<sub>Ac</sub> and cryII<sub>Ab</sub>, controls :
- (1) Cotton bollworm      (2) Corn borer
  - (3) Cotton borer      (4) All the above
- 40.** A nematode *Meloidogyne incognitia* infects the root of tobacco plant and causes a great reduction in yield. A novel strategy was adopted to prevent this infection which was based on the process of :
- (1) DNA interference
  - (2) RNA interference
  - (3) PCR technique
  - (4) DNA test
- 41.** In RNA interference (RNAi) :
- (1) The silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevent translation of the mRNA :
  - (2) The silencing of a specific mRNA due to dsDNA
  - (3) The silencing of a specific mRNA due to tRNA
  - (4) All the above
- 42.** Transgenic animals produces biological product such as  $\alpha$ -1-antitrypsin, which is used to treat :
- (1) Emphysema
  - (2) Cystic fibrosis
  - (3) Phenyl ketonuria
  - (4) Sick cell anaemia

43. The first transgenic cow was 'Rosie', produces :
- (1) Human protein-enriched milk ( $\alpha$ -lactalbumin)
  - (2) Human protein  $\alpha$ -1 antitrypsin riched milk
  - (3) Human protein enriched milk (insulin)
  - (4) All the above
44. Transgenic mice are being developed for use in :
- (1) Testing the safety of polio vaccines before they are used on human
  - (2) Molecular diagnosis of diseases
  - (3) Production of human protein enriched milk
  - (4) Production of human insulin
45. The technique that serves the purpose of early diagnosis of disease or pathogen :
- (1) Recombinant DNA technology
  - (2) Polymerase chain reaction (PCR)
  - (3) Enzyme linked immuno sorbent assay (ELISA)
  - (4) All the above
46. Transgenic tobacco which is developed through RNA interference, prevents the infection of :
- (1) A nematode - *Meloidogyne incognita*
  - (2) A bacterium - *Pseudomonas putida*
  - (3) A fungi - *Tricoderma*
  - (4) An insect
47. The first clinical gene therapy was given to a 4-year old girl with ADA deficiency in :
- |          |          |
|----------|----------|
| (1) 1984 | (2) 1986 |
| (3) 1992 | (4) 1990 |